

# Lab 06

## JavaScript Basics

### Objective:

To familiarize the students with Java Script.

### Activity Outcomes:

After this lab the students should be able to understand

- Java Script basics
- validate the form using Java Script.

### Instructor Note:

As pre-lab activity, read Chapter 8 from the textbook “Web Programming with HTML5, CSS, and JavaScript Pap/Psc John Dean, Year 2018

### 1) Useful Concepts

JavaScript is used in millions of Web pages to improve the design, validate forms, detect browsers, create cookies, and much more. JavaScript is the most popular scripting language on the internet, and works in all major browsers, such as Internet Explorer, Mozilla, Firefox, Netscape, Opera.

How to Put a JavaScript Into an HTML Page?

JavaScript can be embedded into HTML two ways

Inter script: <script>, </script> tags are used to start and end a javascript block

External script: JavaScript is written in a seprate file and included in the HTML file using src attribute of the <script>. Example

```
<script src="myscripts.js"> </script>
```

### Variables in JavaScript:

JavaScript provides numbers, string, boolean, and null variable types. JavaScript is a loosely typed language. var keyword is used to declare a variable. We can declare a variable in JavaScript as

```
var name='Asad';
```

Operats in JavaScript:

- Assignment Operator: =
- Arithmetic Operators: +, -, \*, /, %, ++, --
- Logical Operators: &&, ||, !
- Comparison Operators: ==, ===, !=, !==, <, >, <=, >=

Input/output in JavaScript:

- write(); is used to write on browser
  - o document.write("hello world")
  - o document.write(a)
- prompt(); is used to take input from users
  - o var num = prompt("Please Enter a Number", 0)
- alert(): used to show an alert box

### Defining a function :

```
function functionName([parameters])
{
[statements]
}
```

Conditional statements:

If statement if (condition) statement

or

if(condition)

{ statements } If-else statement if(condition)

{statement} else

{statements}

Loops in JavaScript:

For loop

for(var i=1; i==10; i==)

{

Document.write("hello world")

}

While loop

While(condition)

{

}

Do-while loop

do

{

}

while(condition)

## 2) Solved Lab Activities

| <i>Sr. No</i>     | <i>Time Allocated</i> | <i>Complexity</i> | <i>CLO-Mapping</i> |
|-------------------|-----------------------|-------------------|--------------------|
| <i>Activity 1</i> | <i>10 Minutes</i>     | <i>Medium</i>     | <i>CLO-4</i>       |
| <i>Activity 2</i> | <i>10 Minutes</i>     | <i>Medium</i>     | <i>CLO-4</i>       |
| <i>Activity 3</i> | <i>10 Minutes</i>     | <i>Medium</i>     | <i>CLO-4</i>       |
| <i>Activity 4</i> | <i>10 Minutes</i>     | <i>Medium</i>     | <i>CLO-4</i>       |
| <i>Activity 5</i> | <i>10 Minutes</i>     | <i>Medium</i>     | <i>CLO-4</i>       |

### Activity 1:

*Use JS to take input from user to enter a number and display the entered number by clicking on a button “Tryit”.*

### Solution:

```
<!DOCTYPE html>
<html>
<body>

<h3>A demonstration of how to access a Number field</h3>

<input type="number" id="myNumber" value="2">

<p>Click the button to get the number of the number field.</p>

<button onclick="myFunction()">Try it</button>

<p id="demo"></p>

<script>
function myFunction() {
    var x = document.getElementById("myNumber").value;
    document.getElementById("demo").innerHTML = x;
}
</script>

</body>
</html>
```

## Output:

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>&lt;!DOCTYPE html&gt; &lt;html&gt; &lt;body&gt;  &lt;h3&gt;A demonstration of how to access a Number field&lt;/h3&gt;  &lt;input type="number" id="myNumber" value="2"&gt;  &lt;p&gt;Click the button to get the number of the number field.&lt;/p&gt;  &lt;button onclick="myFunction()"&gt;Try it&lt;/button&gt;  &lt;p id="demo"&gt;&lt;/p&gt;  &lt;script&gt; function myFunction() {   var x = document.getElementById("myNumber").value;   document.getElementById("demo").innerHTML = x; } &lt;/script&gt; &lt;/body&gt; &lt;/html&gt;  </pre> | <p><b>A demonstration of how to access a Number field</b></p> <p>2</p> <p>Click the button to get the number of the number field.</p> <p><input type="button" value="Try it"/></p> <p>2</p> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Activity 2:

*Use JS to get the placeholder text of the search field by Clicking the button Try it*

Input/output and variables:

```
<!DOCTYPE html>
<html>
<body>

<h3>A demonstration of how to access a Search field</h3>

<input type="search" id="mySearch" placeholder="Search for
something..">

<p>Click the button to get the placeholder text of the search
field.</p>

<button onclick="myFunction()">Try it</button>

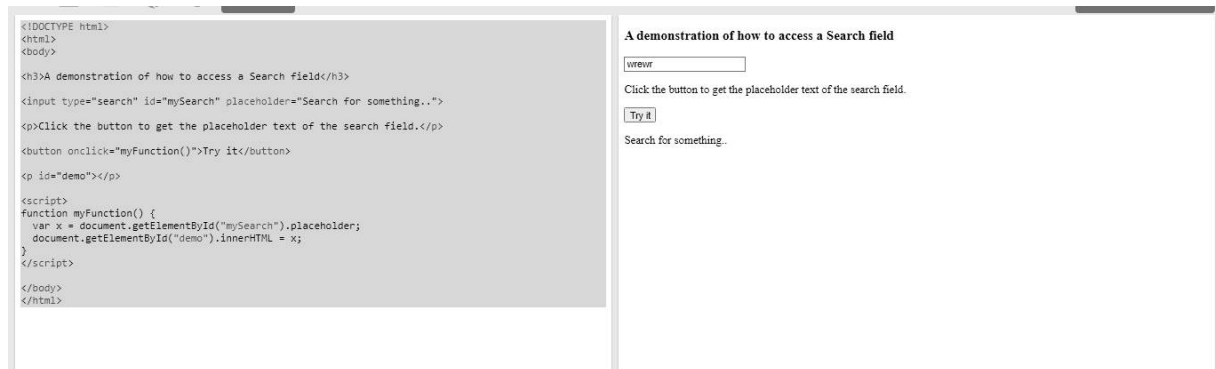
<p id="demo"></p>

<script>
function myFunction() {
  var x = document.getElementById("mySearch").placeholder;
  document.getElementById("demo").innerHTML = x;
}
</script>

</body>
```

```
</html>
```

## Output:



## Activity 3:

*Use JS to write extra large or extra small numbers in exponent or scientific notation.  
e.g  $123e5 = 12300000$  or  $123e-5 = 0.00123$*

### *Input/output and variables:*

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Numbers</h2>

<p>Extra large or extra small numbers can be written with scientific
(exponent) notation:</p>

<p id="demo"></p>

<script>
let x = 123e5;
let y = 123e-5;
document.getElementById("demo").innerHTML = x + "<br>" + y;
</script>

</body>

</html>
```

## Output:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Numbers</h2>

<p>Extra large or extra small numbers can be written with scientific (exponent) notation:</p>

<p id="demo"></p>

<script>
let x = 123e5;
let y = 123e-5;
document.getElementById("demo").innerHTML = x + "<br>" + y;
</script>

</body>
</html>
```

### JavaScript Numbers

Extra large or extra small numbers can be written with scientific (exponent) notation:

12300000  
0.00123

## Activity 4:

*Use JS Date Function to display the date and time on HTML document.*

*Input/output and variables:*

```
<!DOCTYPE html>
<html lang="en-US">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1">
  <title>Today's Date</title>
</head>

<body>

  <script>
    let d = new Date();
    document.body.innerHTML = "<h1>Today's date is " + d + "</h1>"
  </script>

</body>

</html>
```

## Output:



## Activity 5:

*Use JS Input/output, variables, Functions, Conditional statements to create a game. Get a random number using Math.random function in the JS. In the game it asks user to enter a number. If the entered number matches the randomly generated number it generates an alert you won else it displays the actual number.*

### Solution:

```
<html>
<head>

<title>Using If condition</title>
<script language="javascript">
function playGame()
{
var res=Math.round(Math.random()*10)
var num=prompt("Pleae a number",0)
if(num==res)
document.write("You Won")
else
document.write("Your loss, correct Answer is",res)
}
</script>
</head>
<body>
```

```
<p onClick="playGame()">Play the Game</p>
</body>
</html>
```

## Output:



## 3) Graded Lab Tasks

*Note: The instructor can design graded lab activities according to the level of difficult and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.*

### Lab Task 1-3

- Write the javascript code which asks the users to enter a number or zero to end taking the input. When user enters a 0 the program displays the sum.
- Write a javascript function which displays a question in prompt box and gets its answer. If the answer is correct then shows a success message otherwise displays a error message.
- Write javascript code which asks users to enter a number, a message and displays that message for the number of times as entered by the user.

### Lab Task 4:

- Write a JS code to validate the signup form in Lab 3.



# Lab 07

## JavaScript Timers

### Objectives:

To familiarize students with the concepts of JavaScript Animation using JavaScript Timers.

### Activity Outcomes:

After this lab, the students should be able to understand the purpose and

- use of JavaScript Timers.
- Write timer functions in html for creating Website Carousel.

### Instructor Note:

As pre-lab activity, read Chapter 8 from the textbook “Web Programming with HTML5, CSS, and JavaScript Pap/Psc John Dean, Year 2018. And the website [https://www.w3schools.com/js/js\\_timing.asp](https://www.w3schools.com/js/js_timing.asp) for detailed functionalities.

### 1) Useful Concepts

A timer is a function that enables us to execute a function at a particular time.

Using timers, you can delay the execution of code so that it does not get done at the exact moment an event is triggered or the page is loaded. For example, you can use timers to change the advertisement banners on your website at regular intervals, or display a real-time clock, etc. There are two timer functions in JavaScript: `setTimeout()` and `setInterval()`.

#### **`setTimeout()`**

The `setTimeout()` method calls a function or evaluates an expression after a specified number of milliseconds. The function is only executed once. If you need to repeat execution, use the `setInterval()` method. We use the `clearTimeout()` method to prevent the function from running.

Syntax: `setTimeout(function, delay (in milliseconds))`

This function accepts two parameters: a function, which is the function to execute, and an optional delay parameter, which is the number of milliseconds representing the amount of time to wait before executing the function (1 second = 1000 milliseconds).

#### **`clearTimeout()`**

The `clearTimeout()` method clears a timer set with the `setTimeout()` method. The ID value returned by `setTimeout()` is used as the parameter for the `clearTimeout()` method.

```
id_of_settimeout = setTimeout("javascript function", milliseconds);
```

If the function has not already been executed, you will be able to stop the execution by calling the `clearTimeout()` method.

Syntax: `clearTimeout(id_of_settimeout)`

### **setInterval()**

The `setInterval()` function executes a function or specified piece of code repeatedly at fixed time intervals. The `setInterval()` method will continue calling the function until the

`clearInterval()` is called, or the window is closed. The ID value returned by `setInterval()` is used as the parameter for the `clearInterval()` method.

Syntax: `setInterval(function, intervals(in milliseconds))`.

This function also accepts two parameters: a function, which is the function to execute, and interval, which is the number of milliseconds representing the amount of time to wait before executing the function (1 second = 1000 milliseconds).

### **clearInterval()**

The `clearInterval()` method clears a timer set with the `setInterval()` method. The ID value returned by `setInterval()` is used as the parameter for the `clearInterval()` method.

```
id_of_setinterval = setInterval("javascript function", milliseconds);
```

Then you will be able to stop the execution by calling the `clearInterval()` method.

```
clearInterval(id_of_setinterval );
```

## **2) Solved Lab Activities**

| <i>Sr. No</i>     | <i>Time Allocated</i> | <i>Complexity</i> | <i>CLO-Mapping</i> |
|-------------------|-----------------------|-------------------|--------------------|
| <i>Activity 1</i> | <i>20 Minutes</i>     | <i>Medium</i>     | <i>CLO-1</i>       |
| <i>Activity 2</i> | <i>20 Minutes</i>     | <i>Medium</i>     | <i>CLO-1</i>       |
| <i>Activity 3</i> | <i>20 Minutes</i>     | <i>Medium</i>     | <i>CLO-1</i>       |

## Activity 1:

*In first activity of this lab, we will learn how to use `setTimeout()` method to display a text message after a delay of 3 sec.*

### Solution:

```
<!DOCTYPE html>
<html>
<body>

<p>Click the button to wait 3 seconds, then Msg is displayed.</p>
<p id="demo"></p>

<button onclick="myFunction()">Try it</button>

<script>
var myVar;

function myFunction() {
  myVar = setTimeout(MsgFunc, 3000);
}

function MsgFunc() {
  document.getElementById("demo").innerHTML="This text is displayed after three
seconds";
}
</script>
</body>
```

Click the button to wait 3 seconds, then Msg is displayed.

This text is displayed after three seconds

Try it

## Activity 2:

*In second activity of this lab, we will learn how to use `clearTimeout()` method to stop the execution of a function before it is activated.*

### Solution:

```
<!DOCTYPE html>
<html>
<body>

<p>Click the button to wait 3 seconds, then Msg is displayed.</p>
<p id="demo"></p>

<button onclick="start()">Start</button>
<button onclick="stop()">Stop</button>

<script>
var myVar;

function start() {
  myVar = setTimeout(MsgFunc, 3000);
}

function stop() {
  clearTimeout(myVar)
}

function MsgFunc() {
  document.getElementById("demo").innerHTML="This text is displayed after three
seconds";
}
</script>
```

Click the button to wait 3 seconds, then Msg is displayed.

Start Stop

### Activity 3:

*In first activity of this lab, we will learn how to use `setTimeout()` method to display a text message after a delay of 3 sec. In third activity of this lab, we will learn how to use `setInterval()` method for creating a clock which displays current time by updating it against each second.*

### Solution:

```
<!DOCTYPE html>
<html>
<body>

<p>A script on this page starts this clock:</p>
<p id="demo"></p>

<button onclick="myStopFunction()">Stop time</button>

<script>
var myVar = setInterval(myTimer, 1000);

function myTimer() {
    var d = new Date();
    var t = d.toLocaleTimeString();
    document.getElementById("demo").innerHTML = t;
}

function myStopFunction() {
    clearInterval(myVar);
}
</script>

</body>
</html>
```

A script on this page starts this clock:

12:39:28 PM

Stop time

### Activity 3:

*In forth activity of this lab, we will learn how to use `clearInterval()` method for stop the clock at any particular moment within its updating.*

### Solution:

```
<!DOCTYPE html>
<html>
<body>

<p>A script on this page starts this clock:</p>
<p id="demo"></p>

<button onclick="myStopFunction()">Stop time</button>

<script>
var myVar = setInterval(myTimer, 1000);

function myTimer() {
    var d = new Date();
    var t = d.toLocaleTimeString();
    document.getElementById("demo").innerHTML = t;
}

function myStopFunction() {
    clearInterval(myVar);
}
</script>

</body>
</html>
```

A script on this page starts this clock:

12:39:28 PM

Stop time

## 3) Graded Lab Tasks

*Note: The instructor can design graded lab activities according to the level of difficult and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.*

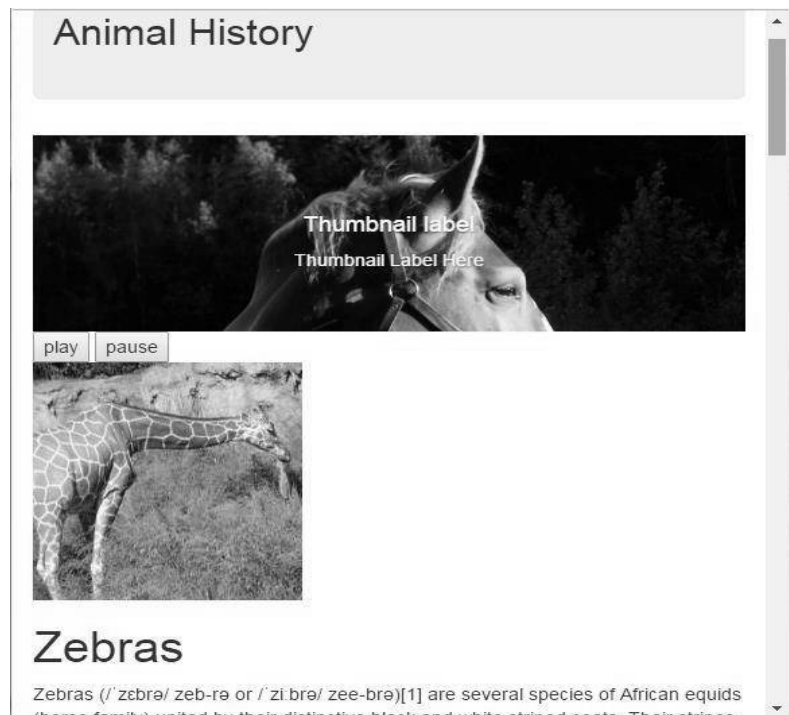
### Lab Task 1

Create a Website Carousel using JavaScript for a responsive animal history web page with its layout for different screen sizes as provided below. Following are the supplementary details about different sections of the page:

- 1- The task should use bootstrap classes to create layout for Animal History banner, Image Slider and Animal description sections of the web page. Students can only use bootstrap.min.css as a third-party CSS library.
- 2- Students are supposed to write a JavaScript code for imageSlider with Play and Pause buttons. Slider should stop when Pause button is pressed and start when Play button is clicked.

3- It is pertinent to note that in descriptive sections of different animals, location of image and description changes with different screen sizes.

1. Extra-small, Small Screen



Medium Screen



## Zebras

Zebras (/ˈzɛbrə/ zeb-rə or /ˈzi brə/ zee-brə)[1] are several species of African equids (horse family) united by their distinctive black and white striped coats. Their stripes come in different patterns, unique to each individual. They are generally social animals that live in small harems to large herds. Unlike their closest relatives, horses and donkeys, zebras have never been truly domesticated.

There are three species of zebras: the plains zebra, the Grévy's zebra and the mountain zebra. The plains zebra and the mountain zebra belong to the subgenus *Hippotigris*, but Grévy's zebra is the sole species of subgenus *Dolichohippus*. The latter resembles an ass, to which it is closely

Large Screen

## Animal History



Thumbnail label

Thumbnail Label Here



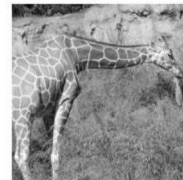
### Zebras

Zebras (/ˈzɛbrə/ zeb-rə or /ˈziːbrə/ zee-brə)[1] are several species of African equids (horse family) united by their distinctive black and white striped coats. Their stripes come in different patterns, unique to each individual. They are generally social animals that live in small harems to large herds. Unlike their closest relatives, horses and donkeys, zebras have never been truly domesticated.

There are three species of zebras: the plains zebra, the Grévy's zebra and the mountain zebra. The plains zebra and the mountain zebra belong to the subgenus *Hippotigris*, but Grévy's zebra is the sole species of subgenus *Dolichohippus*. The latter resembles an ass, to which it is closely related, while the former two are more horse-like. All three belong to the genus *Equus*, along with other living equids.

The unique stripes of zebras make them one of the animals most familiar to people. They occur in a variety of habitats, such as grasslands, savannas, woodlands, thorny scrublands, mountains, and coastal hills. However, various anthropogenic factors have had a severe impact on zebra populations, in particular hunting for skins and habitat destruction. Grévy's zebra and the mountain zebra are endangered. While plains zebras are much more plentiful, one subspecies, the quagga, became extinct in the late 19th century – though there is currently a plan, called the Quagga Project, that aims to breed zebras that are phenotypically similar to the quagga in a process called breeding back.

Zebras evolved among the Old World horses within the last 4 million years. It has been suggested that zebras are polyphyletic and that striped equids evolved more than once. Extensive stripes are posited to have been of little use to equids that live in low densities in deserts (like asses and some horses) or ones that live in colder climates with shaggy coats and annual shading (like some horses).[4] However, molecular evidence supports zebras as a monophyletic lineage



### Leopard





## Lab 08

### JavaScript Form Validation, Regular Expressions

#### Objective:

This lab will introduce you to regular expressions (REGEX) and its practical implementation by demonstrating the use of regular expressions in different views.

#### Activity Outcomes:

After this lab the students should be able to understand Java Script basics and to validate the form using Java Script.

#### This lab teaches you the following topics:

- Purpose and use of different regular expressions
- To write functions in html, validate using regular expressions
- Use of REGEX for validating different inputs (email, password etc).
- Use of REGEX for checking certain words in a string/comment.

#### Instructor Note:

As pre-lab activity, read Chapter 9 from the textbook “Web Programming with HTML5, CSS, and JavaScript Pap/Psc John Dean, Year 2018.

#### 1) Useful Concepts

A regular expression is an object that describes a pattern of characters. Regular expressions are nothing more than a sequence or pattern of characters itself. They provide the foundation for pattern-matching functionality. They are used to perform pattern-matching and "search- and-replace" functions on text. Following link is helpful in creating regular expressions of your requirements.  
[https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide/Regular\\_Expressions](https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide/Regular_Expressions)

#### 2) Solved Lab Activites

| <i>Sr. No</i>     | <i>Time Allocated</i> | <i>Complexity</i> | <i>CLO-Mapping</i> |
|-------------------|-----------------------|-------------------|--------------------|
| <i>Activity 1</i> | <i>20 Minutes</i>     | <i>Medium</i>     | <i>CLO-4</i>       |
| <i>Activity 2</i> | <i>20 Minutes</i>     | <i>Medium</i>     | <i>CLO-4</i>       |
| <i>Activity 3</i> | <i>20 Minutes</i>     | <i>Medium</i>     | <i>CLO-4</i>       |

## Activity 1:

*Develop and demonstrate, using Javascript script, HTML document that collects the user input (the valid format is: A digit from 1 to 4 followed by two upper-case characters followed by two digits followed by two upper-case characters followed by three digits; no embedded spaces allowed) of the user. Event handler must be included for the form element that collects this information to validate the input. Messages in the alert windows must be produced when errors are detected.*

## Solution:

```
1
2 <!DOCTYPE html>
3 <html>
4 <head> <title> Input Validation </title> </head>
5 <body>
6 <form>
7 Note: A digit from 1 to 4, Two upper-case characters, Two digits, Two upper-case characters, Three digits.<br/>
8 User Input: <input type="text" name="user" /> <br/>
9 <input type="button" value="Validate" onclick="valid
10 (user)" />
11 </form>
12
13 <script type="text/javascript">
14 function valid(user)
15 {
16 var pattern1=/^[1-4][A-Z]{2}[0-9]{2}[A-Z]{2}[0-9]{3}$/
17 if(!user.value.match(pattern1)||user.value.length==0)
18 {
19 alert("Invalid User Input!\nEnter a valid input")
20 return false
21 }
22 else alert("Input is valid!")
23 }
24 </script>
25
26 </body>
27 </html>
28
```

## Activity 2:

*Using REGEX to perform character recognition in a string. Also use global(g) case sensitive(i) and multiline(m) search.*

## Solution:

```
1 <!DOCTYPE html>
2 <html>
3 <body>
4
5 <p>Click the button to do a global search for characters NOT inside the brackets [h] in a string.</p>
6
7 <button onclick="myFunction()">Try it</button>
8
9 <p id="demo"></p>
10
11 <script>
12 function myFunction() {
13     var str = "Is this all there is?";
14     var patt1 = /^[^h]/g;
15     var result = str.match(patt1);
16     document.getElementById("demo").innerHTML = result;
17 }
18 </script>
19
20 </body>
21 </html>
```

## Activity 3:

*Validate email field while creating a signup form. Also show the error messages if email is not valid. Use event handler for input field to validate email.*

## Solution:

```
1 <script src="https://code.jquery.com/jquery-3.1.1.min.js" integrity="sha256-hVVnYaiADRTO2PzUGmuLJr8BLUSjGIZsDYGmIJLv2b8="
  crossorigin="anonymous"></script>
2
3 <form method="post">
4     <label for="email" id="email-ariaLabel">Your email address:</label>
5     <input id="email" name="email" type="email" class="required"/>
6     <span id="error">Not a valid email address</span>
7     <span id="success">A valid email address!</span>
8 </form>
9
10 <script>
11 $(document).ready(function(){
12     $("#success").hide();
13     $("#error").hide();
14     $("#email").keyup(function () {
15         var email = $("#email").val();
16         // i Perform case-insensitive matching
17         // g Perform a global match (find all matches rather than stopping after the first match)
18         // m Perform multiline matching
19         var re = /[A-Z0-9._%+-]+@[A-Z0-9.-]+\.[A-Z]{2,4}/igm;
20         if (re.test(email)) {
21             $('#success').show();
22             $('#error').hide();
23         } else {
24             $('#error').show();
25             $('#success').hide();
26         }
27     });
28 });
29 </script>
```

### 3) Graded Lab Tasks

*Note: The instructor can design graded lab activities according to the level of difficult and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.*

#### Lab Task 1

*Validate password field while creating signup form. Ask the user to set a strong password by using at least 1 uppercase letter, 1 lowercase letter, 1 digit and 1 ASCII character show the error messages if password doesn't contain any of these expressions.*